



Empirical Analysis of the Double Descent Phenomenon

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Abstract

The classical understanding of machine learning is centered on the bias-variance trade-off, which typically suggests a U-shaped risk curve where increasing model capacity leads to overfitting after a certain point [1, 2]. However, modern practice involving deep learning shows that very rich models can generalize well even when they perfectly fit or interpolate noisy training data [1].

This project explores the “double descent” phenomenon, a term coined to describe a unified performance curve [1]. This curve reconciles classical theory with modern practice by showing that, beyond the interpolation threshold, that is the point in which the number of parameters of the model is equal to the number of used training samples, increasing model capacity (over-parameterization) can actually lead to a second decrease in test risk. Our goal is to empirically reproduce this behavior using a linear regression framework.

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Chapter 1

Dataset Generation and Methodology

1.1 Generating the Synthetic Dataset

To analyze the risk curve, we generate a synthetic dataset in which the input matrix $X \in \mathbb{R}^{n \times d}$ is sampled from a standard normal distribution [1] or it is determined using a feature decay approach. The ground truth is defined by a weight vector w^* . The ultimate benchmark for any predictor is the **Bayes optimal predictor** (f^*), which represents the theoretical minimum risk achievable for a given distribution. For the squared loss, $f^*(x)$ is defined as the expected value of the label conditioned on the instance: $f^*(x) = \mathbb{E}[Y|X = x]$. The corresponding **Bayes risk** is the expected conditional variance of the label: $\mathbb{E}[\text{Var}[Y|X]]$ [2]. The synthetic setup is designed such that the noiseless labels represent this optimal signal, while the added Gaussian noise defines the inherent Bayes risk of the problem. The resulting “clean” labels, $y_{\text{clean}} = Xw^*$, represent the **Bayes optimal predictor** for the squared loss.

1.1.1 Feature decay

This deterministic approach was used to concentrate the signal within the initial features, effectively countering the high initial **approximation error** (or bias). In this context, bias is defined as the difference between the test error of the best predictor in the class $\mathcal{H}$ and the Bayes risk. By structuring w^* this way, the simulation highlights the “sweet spot” of the classical regime, which emerges from a rapid decrease in bias followed by a sharp increase in estimation error (or variance). The Variance here represents the difference between the predictor’s test error and the test error of the best predictor in the class, as shown in the following bias-variance decomposition of the risk:

$$\ell_D(h_S) = \underbrace{\ell_D(h_S) - \ell_D(h^*)}_{\text{Estimation error (Varianza)}} + \underbrace{\ell_D(h^*) - \ell_D(f^*)}_{\text{Approximation error (Bias)}} + \underbrace{\ell_D(f^*)}_{\text{Bayes risk}}$$

The rise in **variance** near the interpolation threshold is driven by the instability of least squares, a technical issue arising from the inversion of the matrix $X^\top X$ when it becomes nearly singular. Furthermore, this method illustrates the second descent in the modern regime, where the test risk decreases again. This phenomenon is attributed to the **minimum norm solution**; since the model has already captured most of the useful signal, it focuses solely on optimizing the smoothness and regularity of the interpolating function. A similar methodological choice is found in the synthetic model proposed by the paper “Reconciling modern machine learning practice and the bias-variance trade-off” [1], where target coefficients decay according to a polynomial law.

1.2 Adding Noise and Signal-to-Noise Ratio (SNR)

Following the synthetic model approach, we introduce stochastic noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ to the labels. The noise level is controlled via the **Signal-to-Noise Ratio (SNR)**, which compares the signal power to the noise variance [1]. For our experiments, we set $SNR = 2$.

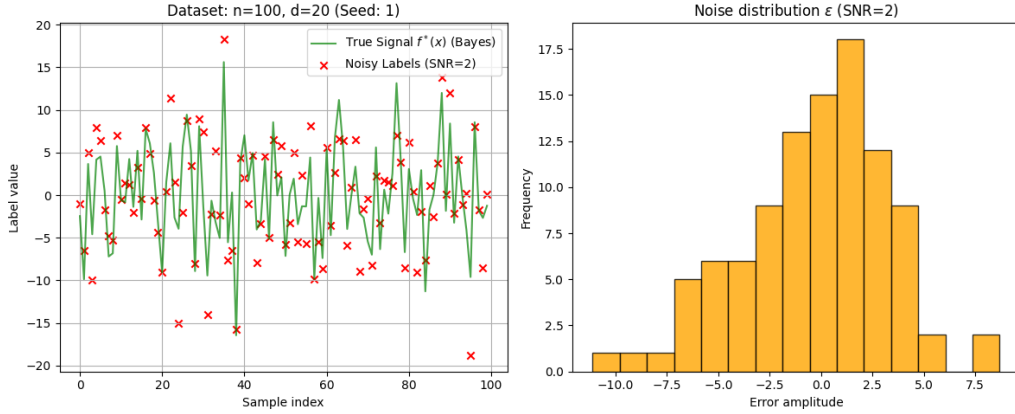


Figure 1.1: Comparison of signal vs. noisy labels ($SNR = 2$).

1.3 Training methodology

The analytical phase involved a systematic evaluation of several regression models: **Ordinary Least Squares (OLS)** utilizing the Moore-Penrose pseudoinverse, **Ridge Regression**, **Gradient Descent (GD)**, and a **hybrid implementation** that applies standard OLS within the classical under-parameterized regime and transitions to Ridge Regression in the modern over-parameterized regime. To ensure statistical stability and reduce variance in the risk estimations, all performance metrics were computed by averaging the results over **20 independent trials**. The simulations were conducted across **four distinct data configurations** to observe the robustness of the Double Descent phenomenon:

1. Datasets with and without feature decay (deterministic weight structuring).
2. Datasets with and without label noise (stochastic perturbations).

The **primary focus** of this study is the behavior of the OLS, Ridge regression, and Gradient Descent models when applied to datasets of **100 samples characterized by feature decay and the presence of noise**. This specific configuration is essential for highlighting the classical "U-shaped" risk curve, as it allows for a clear decomposition of the approximation error (bias)—which drops rapidly due to signal concentration—and the estimation error (variance), which escalates near the interpolation threshold due to the mathematical instability of the least squares solution.

Chapter 2

Model Training

The training phase of this project is rooted in the principle of **Empirical Risk Minimization (ERM)**. For the regression tasks at hand, we utilize the **squared loss** function, defined as $\ell(y, \hat{y}) = (y - \hat{y})^2$ [2]. The objective of the learning algorithm is to find a weight vector $w \in \mathbb{R}^d$ that minimizes the training error on a dataset S of size n :

$$\ell_S(h) = \frac{1}{n} \sum_{i=1}^n (w^\top x_i - y_i)^2 \quad (2.1)$$

In matrix notation, this corresponds to minimizing the objective function $\|Xw - y\|^2$, where $X \in \mathbb{R}^{n \times d}$ is the design matrix and $y \in \mathbb{R}^n$ is the vector of labels [1, 2].

2.1 Ordinary Least Squares and the Minimum Norm Solution

The performance of the **Ordinary Least Squares (OLS)** predictor is central to our study of the double descent phenomenon, as it behaves differently depending on the relationship between the number of samples n and the number of parameters d [1].

2.1.1 The Under-parameterized Regime ($d < n$)

In the classical regime, where the number of parameters d is less than the number of samples n , the matrix $X^\top X$ is typically non-singular and invertible. The unique ERM solution is given by the standard closed-form expression:

$$w = (X^\top X)^{-1} X^\top y \quad (2.2)$$

As discussed in [2], this regime is characterized by the traditional **U-shaped risk curve**. In this context, increasing model capacity beyond a certain "sweet spot" leads to overfitting, where the model begins to fit the noise in the training data, causing the test risk to rise [1].

2.1.2 The Over-parameterized Regime ($d > n$)

When the number of parameters exceeds the **interpolation threshold** ($d = n$), the model class is sufficiently rich to achieve zero training risk, effectively interpolating the training data [1]. In this scenario, the system $Xw = y$ is under-determined and admits an infinite number of solutions.

To identify a unique predictor, we follow the approach proposed by [1] and select the solution with the **minimum ℓ_2 norm**. This choice acts as an **inductive bias** that favors the "smoothest" function compatible with the training data. This preference for simplicity among interpolating models is a mathematical application of **Occam's Razor**.

2.1.3 Implementation via Pseudoinverse

In our numerical implementation, we calculate the minimum norm solution using the **Moore-Penrose pseudoinverse** (X^+). This approach, described in [2], provides a stable way to compute $w = X^+y$ in both regimes. By using the pseudoinverse, we can observe the second descent in the risk curve, as increasing d beyond n allows the model to find interpolating solutions with progressively smaller norms, thereby improving generalization [1].

2.1.4 Results

- **Underparameterized Regime:** Ordinary Least Squares (OLS) behaves according to classical statistical expectations. Initially, the test error is high due to *underfitting*, as the model lacks the capacity to capture the underlying data structure. As model complexity increases, the *approximation error* decreases, leading to the *sweet spot* where the risk is minimized. Beyond this point, the test error rises again as the *estimation error* increases when approaching the *interpolation threshold*.
- **Overparameterized Regime:** Contrary to classical intuition, the test error exhibits a steady decline. This occurs because the model identifies the *minimum-norm solution*, which effectively reduces the estimation error and allows the risk to converge toward the *Bayes risk*.

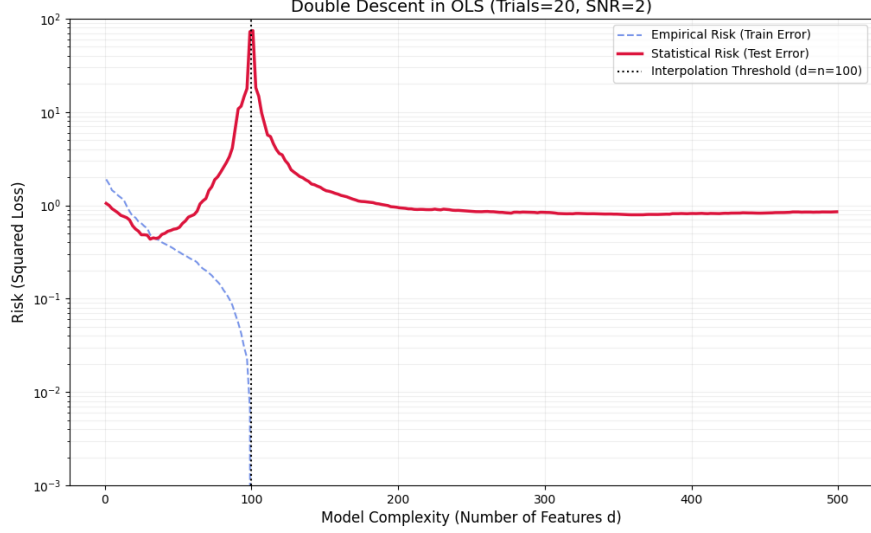


Figure 2.1: OLS with pseudoinverse($SNR = 2$).

- **Training Error:** As expected, the training error drops sharply toward zero at the *interpolation point*. At this threshold, the model possesses sufficient degrees of freedom to perfectly fit the training data by utilizing the full dimensionality of the feature space.

2.2 Ridge Regression

The use of **Ridge Regression** provides several mathematical and computational advantages over standard Ordinary Least Squares (OLS), particularly when dealing with high-dimensional datasets and the interpolation threshold, by introducing a regularization parameter added to the $X^T X$ matrix.

$$w = (\alpha I + X^T X)^{-1} X^T y \quad (2.3)$$

2.2.1 Benefits

The following section outlines these benefits in formal terms.

- **Guaranteed Matrix Invertibility:** the standard OLS solution, defined as $w = (X^T X)^{-1} X^T y$, requires the matrix $X^T X$ to be non-singular. In practice, when the number of features d equals or exceeds the number of samples n , or when features are highly correlated, $X^T X$

becomes singular or nearly singular, making its inversion impossible or numerically unstable. Ridge regression resolves this by adding a regularization term αI (where $\alpha > 0$ and I is the identity matrix). If $\lambda_1, \dots, \lambda_d$ are the eigenvalues of $X^\top X$, the eigenvalues of the regularized matrix $(\alpha I + X^\top X)$ are $\alpha + \lambda_i$. Since $\alpha > 0$ and $\lambda_i \geq 0$, all eigenvalues are strictly positive (not zero), ensuring the matrix is always invertible.

- **Reduction of Estimation Error (Variance):** When the data matrix is nearly singular, the OLS predictor becomes highly sensitive to small perturbations in the training set, leading to an explosion in *estimation error* (variance). By introducing the penalty term $\alpha \|w\|^2$, Ridge regression stabilizes the predictor. Although this increases the *approximation error* (bias), the significant reduction in variance often results in a lower total statistical risk.
- **Numerical Conditioning and the Modern Regime:** From a computational standpoint, Ridge regression provides a more robust way to approach the *minimum-norm solution* used in the modern over-parameterized regime. As $\alpha \rightarrow 0^+$, the Ridge solution converges to the Moore-Penrose pseudoinverse solution (X^+y) , which identifies the "smoothest" interpolant among the infinite possible solutions when $d > n$. Using a small but non-zero α (e.g., 10^{-6}) avoids the extreme numerical instabilities encountered exactly at the interpolation threshold while still allowing for the observation of the *Double Descent* phenomenon.
- **Effective Complexity Control:** The parameter α serves as a continuous "tuning knob" for model capacity. While standard OLS only allows for discrete changes in capacity (by adding or removing features), Ridge regression allows for smooth transitions between a highly constrained model ($\alpha \rightarrow \infty$) and a perfectly interpolating one ($\alpha \rightarrow 0$), enabling finer hyperparameter optimization through techniques like cross-validation.

2.2.2 Results

As shown in 2.2, when applying Ridge regression instead of Ordinary Least Squares (OLS), the double descent behavior is still present but becomes significantly smoother around the interpolation threshold. In particular, the

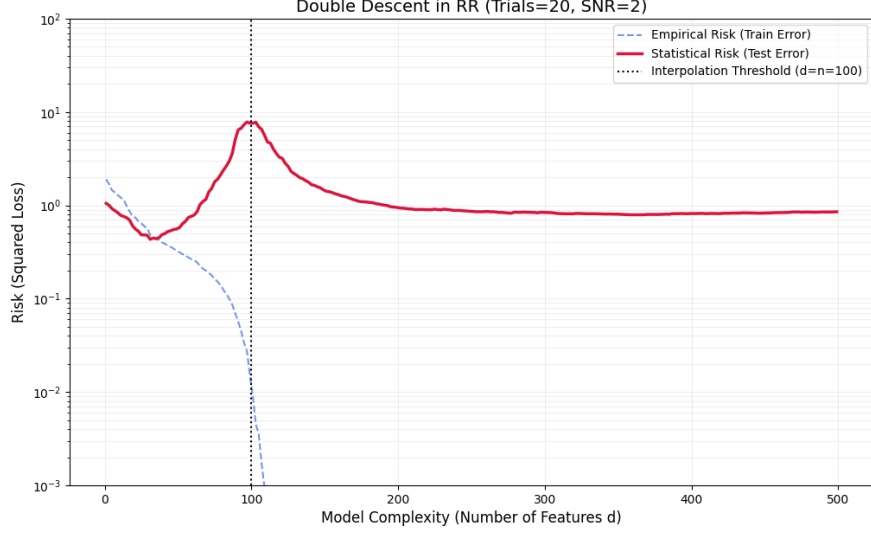


Figure 2.2: Ridge regression with $\alpha=0.1$ ($SNR = 2$).

sharp peak observed in the OLS case at $d \approx n$ is attenuated due to the presence of the regularization term αI , which improves the conditioning of the matrix $X^\top X$ by shifting its eigenvalues away from zero. Unlike OLS, where the training error reaches zero exactly at the interpolation threshold ($d \geq n$), Ridge regression does not interpolate the training data. The regularization term introduces an **explicit bias** that prevents perfect fitting. Consequently, the training error decreases more gradually and remains strictly greater than zero even in the overparameterized regime. This persistent error is a direct consequence of the bias induced by regularization, which trades off variance reduction for increased bias.

2.3 Gradient Descent

At each iteration, the parameter vector w is updated in the opposite direction of the gradient of the loss function:

$$w_{t+1} = w_t - \eta \nabla \ell(w_t) \quad (2.4)$$

where the gradient is given by:

$$\nabla \ell = \frac{2}{n} X^\top (Xw - y) \quad (2.5)$$

The factor $1/n$ ensures that the gradient magnitude is independent of the number of training samples, improving numerical stability and making the learning rate transferable across different dataset sizes.

2.3.1 Hyperparameters

The behavior of Gradient Descent is controlled by the following hyperparameters:

- **Learning rate** $\eta = 10^{-3}$, which determines the step size of each update.
- **Maximum number of iterations** $T_{\max} = 2500$, which sets an upper bound on the optimization time.
- **Tolerance** $\text{tol} = 10^{-8}$, used for early stopping based on the convergence of the training loss.

Early stopping is triggered when the absolute improvement in training loss satisfies:

$$|\ell(t) - \ell^{(t-1)}| < \text{tol} \quad (2.6)$$

2.3.2 Results

The use of Gradient Descent introduces an implicit bias depending on the number of optimization iterations. In the underparameterized regime ($d < n$), the algorithm converges to a stable solution that progressively reduces both training and test error as model complexity increases.

At the interpolation threshold ($d \approx n$), the optimization landscape becomes ill-conditioned due to the presence of small singular values in X . This leads to increased sensitivity of the learned parameters, resulting in a peak in the test error.

In the overparameterized regime ($d > n$), the behavior of Gradient Descent depends on the convergence dynamics. When allowed to converge fully, GD approaches the minimum-norm solution, leading to a reduction in test error as the model complexity increases further. However, due to early stopping based on the tolerance criterion, the optimization may halt before fully fitting all directions in parameter space. This induces an additional bias, preventing the training error from reaching exactly zero and smoothing the overall double descent curve.

Overall, Gradient Descent acts as an implicit regularizer, where the number of iterations and convergence tolerance control the trade-off between bias

and variance, thereby shaping the observed double descent phenomenon in both training and test errors.

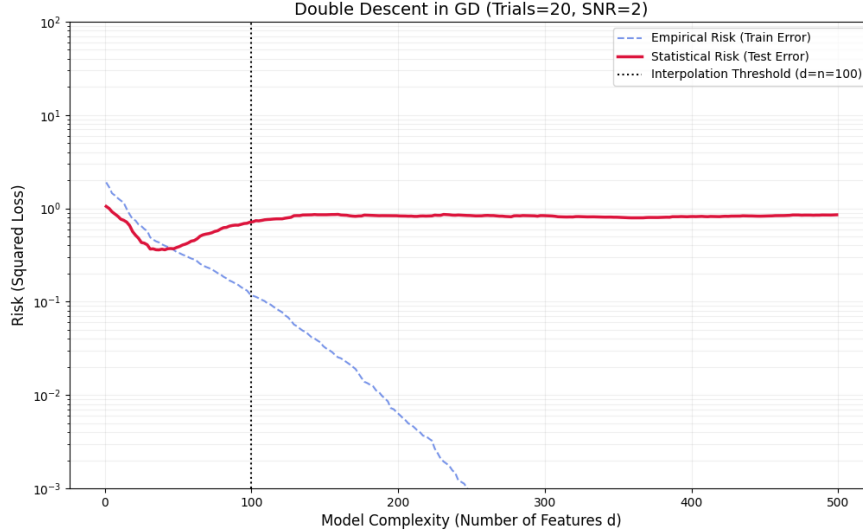


Figure 2.3: Gradient descent with $\eta = 10^{-3}$, max iterations= 2500 and tolerance = 10^{-8} .

2.4 OLS in the Noiseless Setting

When Ordinary Least Squares (OLS) is applied to a noiseless training set, the model exhibits distinct behaviors across different regimes of model complexity.

Underparameterized regime ($d < n$). In this regime, the model has insufficient capacity to represent the true underlying function exactly. However, as the number of features increases, the **approximation error decreases monotonically**. In the absence of noise, the training error rapidly converges to zero. As expected, without noise the model is able to achieve almost zero test error as soon as it has enough feature to reduce the approximation error.

Overparameterized regime ($d > n$). In this regime, infinitely many solutions interpolate the training data. OLS selects the minimum-norm interpolating solution via the Moore-Penrose pseudoinverse. However, this

solution may exhibit high variance due to its dependence on directions associated with small singular values of X . As a result, despite perfect interpolation of the training set, the test error increases significantly and reflects the instability of the chosen interpolating solution rather than the presence of noise.

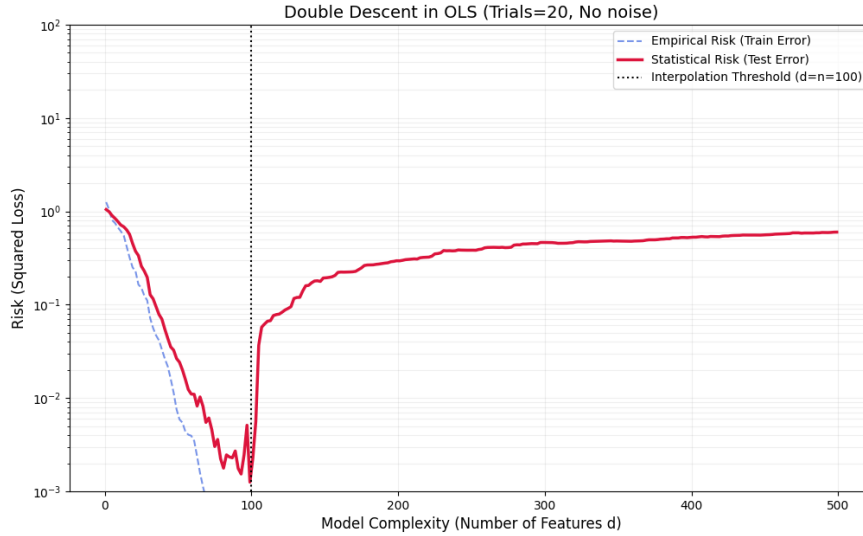


Figure 2.4: OLS in noiseless training labels setting.

2.5 Gradient Descent in the Noiseless Setting

In the noiseless setting, the empirical risk admits a global minimum with zero training error. When GD is run until convergence, it converges to the minimum-norm interpolating solution. An important aspect of GD is that different directions in parameter space are learned at different rates. Directions corresponding to **large** singular values of X are learned **quickly**, while those associated with **small** singular values are learned **slowly**. Therefore, if GD is stopped early (**early stopping**), it implicitly avoids fitting the unstable directions, acting as a form of **regularization** and potentially improving generalization.

Even in the **absence of noise**, the test error exhibits a non-monotonic behavior due to the interaction between model complexity and the geometry of the data matrix. The increase in test error near the interpolation threshold

is not due to overfitting noise, but rather to the instability induced by **small singular values** of X .

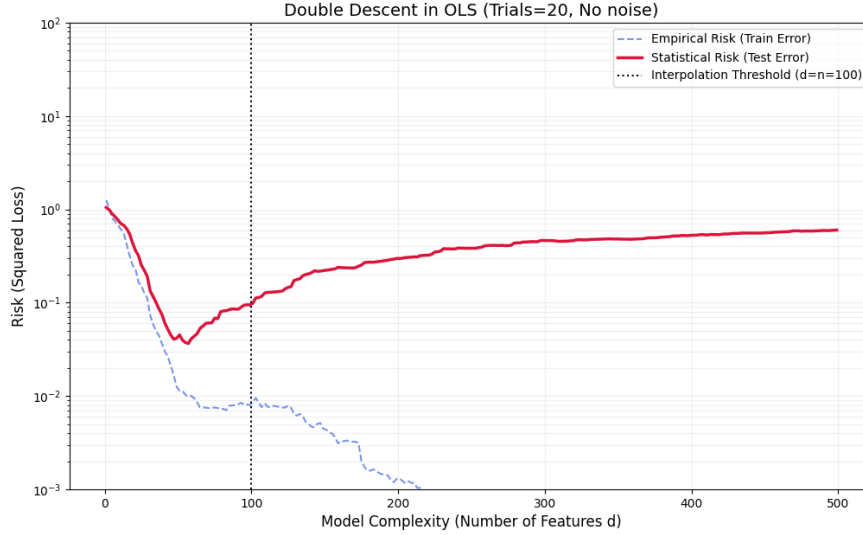


Figure 2.5: GD in noiseless training labels setting.

2.6 Appendix

2.6.1 Reproducibility and Parameter Selection

We use a fixed `seed` with the `numpy.random.default_rng(seed)` generator to ensure that X , w^* , and ϵ are identical across different runs. This allows us to isolate the effect of changing the number of features d relative to the number of samples n . The seed used to plot the examples is `seed = 1` for all pseudo-number generation.

2.6.2 Statement of Authorship

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This

assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

Bibliography

- [1] M. Belkin, D. Hsu, S. Ma, and S. Mandal. *Reconciling modern machine learning practice and the bias-variance trade-off*. PNAS, 2019.
- [2] N. Cesa-Bianchi. *Statistical Methods for Machine Learning*. Lecture Notes, 2026.